

## Interfacial Electron Modulation with Ionic Liquids: Dual Optimization of CO<sub>2</sub> Confinement and Charge Transfer for Enhanced Electoreduction on Cu

Chuanhui Wang, Mingyue Qiu, Guizeng Liang, Hui Yan, Jiamin Ma, Lijuan Shi,\* and Congyun Zhang\*



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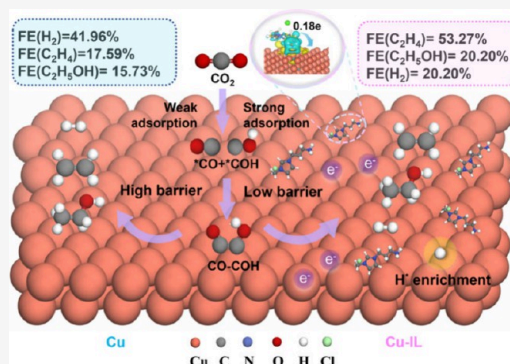


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**ABSTRACT:** The high-rate electrocatalytic CO<sub>2</sub> reduction reaction (CO<sub>2</sub>RR) to afford multicarbon products (C<sub>2</sub>) holds transformative potential for advancing sustainable energy systems. Ionic liquids (ILs) have emerged as dynamic modulators to promote the C<sub>2</sub> pathway, yet the underlying atomistic mechanisms of ILs in modulating this CO<sub>2</sub>RR process remain fundamentally unclear. Here, by integrating molecular dynamics (MD) simulations, density functional theory (DFT) calculations, and experimental validations, we systematically elucidate the critical influence of 1-(3-aminopropyl)-3-methylimidazole chloride ILs in optimizing the CO<sub>2</sub>RR pathway on Cu surfaces through electronic structure engineering. MD simulations demonstrate that ILs establish a CO<sub>2</sub>-enriched interfacial microenvironment that restricts bulk-phase CO<sub>2</sub> diffusion through the confinement effect. Electron structure analyses reveal that ILs synergistically enhance interfacial electron accumulation and directional charge transfer for adsorbed CO<sub>2</sub> and key intermediates (\*CO, \*COH, \*CHO, and \*C), collectively stabilizing them through IL-induced strengthening of Cu–C bonding. More importantly, the introduction of ILs dramatically reduces the activation barrier of the rate-determining C–C coupling step and thermodynamically favors the CO<sub>2</sub>RR to C<sub>2</sub>H<sub>4</sub> and C<sub>2</sub>H<sub>5</sub>OH pathways through atom orbital hybridization. Additionally, the ILs not only enhance the CO<sub>2</sub>RR but also suppress the hydrogen evolution reaction (HER) through proton confinement. This work provides molecular-level insights into the dynamic role of ILs in optimizing CO<sub>2</sub>RR processes and offers a foundation for designing advanced IL-mediated electrocatalytic systems.



### INTRODUCTION

The electrochemical CO<sub>2</sub> reduction reaction (CO<sub>2</sub>RR) has emerged as a promising strategy for mitigating atmospheric CO<sub>2</sub> levels while establishing carbon-neutral energy cycles.<sup>1,2</sup> High-rate CO<sub>2</sub> electrolysis to yield high-value-added multicarbon (C<sub>2+</sub>) products offers both environmental and economic benefits by simultaneously utilizing waste carbon and producing industrially essential feedstocks.<sup>3–5</sup> Among these, ethylene (C<sub>2</sub>H<sub>4</sub>) and ethanol (C<sub>2</sub>H<sub>5</sub>OH), which can serve as fundamental building blocks for the synthesis of long-chain hydrocarbons, oxygenates, and polymers, have garnered substantial attention due to their widespread industrial application and energy storage potential.<sup>6,7</sup> Nevertheless, achieving simultaneously high activity and selectivity toward specific C<sub>2</sub> products remains challenging: (i) thermodynamic competition from the hydrogen evolution reaction (HER),<sup>8</sup> (ii) limited CO<sub>2</sub> solubility, high overpotentials, and sluggish kinetics of C–C coupling steps governed by high activation barriers,<sup>9,10</sup> and (iii) insufficient stabilization of critical \*CO intermediates during rate-determining surface processes.<sup>11</sup>

Copper (Cu)-based catalysts have emerged as prominent candidates for the electrochemical CO<sub>2</sub>RR due to their moderate adsorption energy for critical intermediates involved

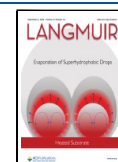
in C<sub>2+</sub> product formation.<sup>12–14</sup> This intrinsic property enables the selective conversion of CO<sub>2</sub> into value-added hydrocarbons while maintaining thermodynamic feasibility.<sup>15</sup> Recent advances in catalytic performance optimization focus on two synergistic strategies: (i) precision design of catalytic microenvironments<sup>16,17</sup> and (ii) microstructural engineering of a copper-based electrode.<sup>18–20</sup> The rational design of an electrochemical interface using polymeric binders or functional additives,<sup>21</sup> such as polyaniline,<sup>22</sup> poly(acrylic acid) derivatives,<sup>23</sup> 1-*n*-alkylimidazolium ionomers,<sup>24</sup> aryl diazonium salts,<sup>25</sup> and ionic liquids (ILs),<sup>26–29</sup> has been demonstrated to enhance C<sub>2+</sub> activity and selectivity. Among these, IL-modified systems exhibit exceptional promise due to their uniquely customizable physicochemical properties, including high electrochemical stability, superior ionic conductivity, and CO<sub>2</sub> solubility capacities, which exceed those of aqueous

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electrolytes by 2–3 orders of magnitude. Additionally, the adjustable hydrophobicity of ILs can enhance the CO<sub>2</sub> affinity and suppress the competing side reaction, HER.<sup>30,31</sup> These distinctive characteristics enable ILs as dual-functional interface modulators, capable of simultaneously serving as structural templates and dynamic reaction mediators.<sup>32,33</sup> For instance, ILs can establish structured electric double layers through specific interactions with reactive intermediates, thereby creating favorable interfacial environments that alter reaction pathways and energetics.<sup>34,35</sup> The imidazolium (Im)-based ILs can interact with reactive intermediates of the CO<sub>2</sub>RR through formation of an Im–CO<sub>2</sub> adduct,<sup>36</sup> which can significantly facilitate the adsorption and activation of CO<sub>2</sub> molecules, thereby decreasing the overpotential of the reaction and effectively suppressing the HER. Theoretical simulations confirm that these interactions alter interfacial energetics, as evidenced by reduced Gibbs free energy barriers and enhanced adsorption energies for critical intermediates. Concurrently, microstructural engineering of Cu-based catalysts has advanced through IL-assisted synthesis.<sup>37</sup> For instance, the ionic liquid (1-octyl-3-methylimidazolium chloride ([Omim]Cl)), serves as a bifunctional agent to fabricate Cu<sub>2</sub>O nanoparticles with rough surfaces and oxygen vacancies, which favor the C<sub>2+</sub> process.<sup>38</sup>

Despite significant advancements in the IL-mediated electrochemical CO<sub>2</sub>RR, the atomistic-level mechanism is still ambiguous. While their ability to modulate microenvironments has been acknowledged, the precise pathways by which ILs influence intermediate stabilization, interfacial charge distribution, and dynamic HER suppression remain unresolved. Density functional theory (DFT)-guided mechanistic insights have proven to be indispensable in resolving the electronic interactions of ILs with catalytic surfaces and reactive intermediates.<sup>39</sup> Recent work employed hybrid functional DFT calculations to reveal how imidazolium-based ILs preferentially stabilize \*COOH intermediates through noncovalent C–H...O interactions, thereby alleviating the thermodynamic bottleneck for CO formation.<sup>40</sup> Complementary DFT simulations further demonstrated room-temperature ionic liquids (RTILs) reduce the thermodynamic barrier for the rate-limiting \*COOH formation step by restructuring interfacial electric double layers, modulating solvation dynamics, and suppressing proton diffusion.<sup>33</sup> Resolving uncertainties in ILs' molecular-level interactions with macro-scale outcomes requires advanced in situ/operando characterization coupled with multiscale modeling to unravel the transient interfacial processes governing CO<sub>2</sub>RR pathways. Addressing these challenges will accelerate the rational design of IL-integrated catalytic systems for scalable CO<sub>2</sub> to fuel conversion.

In this work, we systematically elucidate the atomic-scale mechanism for the influence of the IL-modulated catalytic microenvironment on the CO<sub>2</sub>RR to C<sub>2</sub> conversation at the Cu (1 0 0) interface. By integrating molecular dynamics (MD) simulations, first-principles DFT calculations, in situ spectroscopic evidence, and electrochemical analysis, our findings reveal that ILs simultaneously engineer thermodynamic stabilization and kinetic acceleration through precisely tailored interfacial microenvironments. Thermodynamically, the IL-induced confinement effect creates a CO<sub>2</sub>-enriched interfacial layer, which can enhance CO<sub>2</sub> adsorption and stabilize key intermediates during the CO<sub>2</sub>RR. Kinetically, quantum-level electronic modulation via IL-derived charge redistribution and

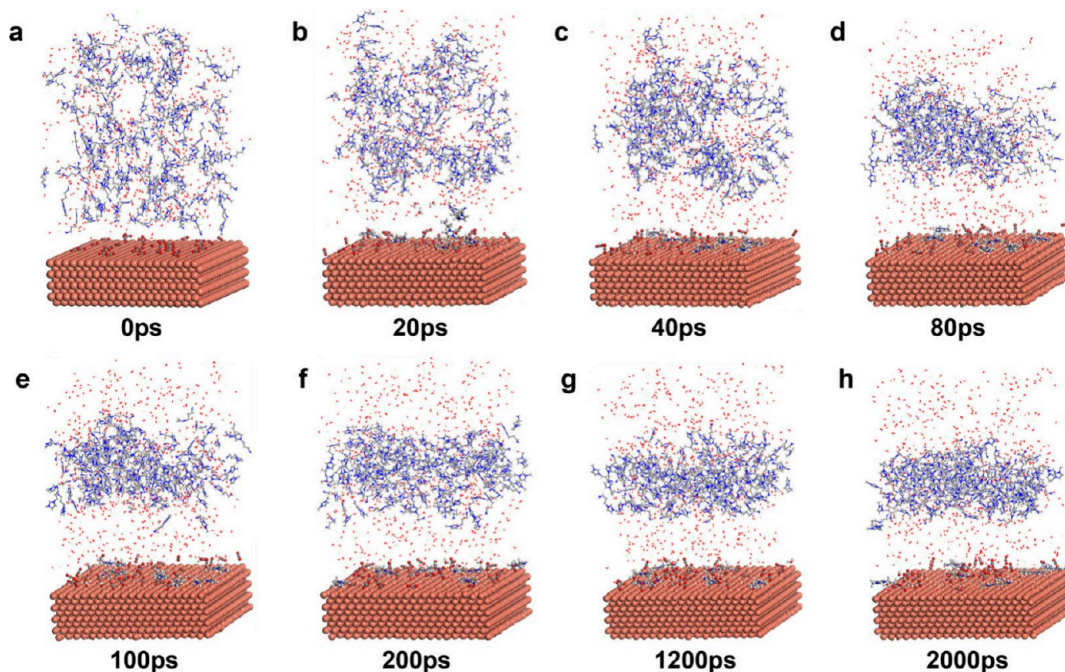
transfer could upshift the d-band center of Cu toward the Fermi level, strengthen Cu–C/O interactions, reduce the energy barrier for the rate-determining step, and facilitate C–C coupling. In particular, the IL-induced microenvironment not only promotes CO<sub>2</sub> to C<sub>2</sub> conversion but also significantly suppresses the competing hydrogen evolution reaction through proton confinement, as confirmed by experimental characterizations. Collectively, these theoretical and experimental investigations establish a universal design paradigm for electrolyte-engineered electrocatalysts, in which targeted tuning of interfacial charge transfer and confinement effects can concurrently optimize the reactant concentration, intermediate stabilization, and transition state energetics. The robust correlation among DFT-predicted adsorption trends, spectroscopic intermediate fingerprints, and electrochemical selectivity metrics underscores the indispensability of theoretical modeling in unraveling complex electrocatalytic mechanisms. By demonstrating how ILs serve as both electronic modulators and interfacial architects, this work provides a quantum chemistry-guided concept for advancing renewable fuel synthesis, a critical step toward closing the carbon cycle in sustainable energy systems.

## ■ EXPERIMENTAL SECTION

**MD Simulation Details.** In this study, we conducted MD simulations to systematically investigate the role of ILs in enhancing CO<sub>2</sub> adsorption and diffusion behavior at the gas–liquid–solid three-phase interface on a Cu (1 0 0) surface. All calculations were performed by using the Forcite module in the Materials Studio 2020 software package. All atomic interactions, including bonded and nonbonded interactions, were modeled using the COMPASS III force field. Van der Waals interactions were calculated using the atom-based method with a cutoff radius of 15.5 Å, while long-range electrostatic interactions were evaluated using the Ewald summation method with an accuracy of  $1 \times 10^{-5}$  kcal/mol. Two representative simulation models were constructed to simulate the IL/water mixed electrolytes: one containing 3200 Cu atoms, 1280 H<sub>2</sub>O molecules, 64 CO<sub>2</sub> molecules, and 320 ILs (here and in subsequent sections, ILs refers to 1-(3-aminopropyl)-3-methylimidazole chloride ionic liquids) and the other system being the same except without ILs for comparison. The initial model was pre-relaxed to the convergence threshold (energy of  $<1 \times 10^{-4}$  kcal/mol, force of  $<0.005$  kcal mol<sup>−1</sup> Å<sup>−1</sup>) through geometry optimization. During the equilibrium stage, the NPT ensemble was employed (constant temperature and constant pressure) for 1 ns, wherein the temperature was controlled by the Andersen thermostat at 300 K and the pressure was maintained at 1 atm through the Berendsen method. After equilibration, production simulations were conducted under the NVT ensemble for 2 ns, with a step size of 1 fs. Trajectory data were recorded every 1 ps. Postsimulation analyses, including the mean square displacement (MSD), radial distribution functions (RDFs), and density statistics, were conducted through the Forcite Analysis tool.

**DFT Simulation Details.** The DFT calculations for the CO<sub>2</sub>RR on the Cu surface and 1-(3-aminopropyl)-3-methylimidazole chloride ionic liquid-modified Cu (Cu–IL) interface were performed using the Vienna Ab initio Simulation Package (VASP).<sup>41,42</sup> The projector-augmented-wave (PAW)<sup>43,44</sup> method was employed to describe the interactions between ions and electrons, and the generalized gradient approximation (GGA) with the Perdew–Burke–Ernzerhof (PBE) functional was used to calculate the energy and electronic structure properties of the materials.<sup>45</sup> A  $6 \times 6$  supercell was used for the Cu (1 0 0) surface. To avoid interactions between periodic units during structural relaxation, a vacuum space of  $>20$  Å was set. During optimization, the bottom two layers of copper atoms were fixed while the other atoms were allowed to relax. To ensure the convergence of the total energy, the kinetic energy cutoff for the plane-wave expansion was set to 500 eV, and all calculations employed a





**Figure 1.** Representative snapshots of CO<sub>2</sub> adsorption in the IL-modified Cu system over time. Panels a–h represent the configurations at 0, 20, 40, 80, 100, 200, 1200, and 2000 ps, respectively.

Monkhorst–Pack  $2 \times 2 \times 1$   $k$ -point mesh. The convergence criteria for forces and energy were set to 0.01 eV/Å and  $10^{-5}$  eV, respectively. In frequency calculations, the energy convergence criterion was set to  $1 \times 10^{-7}$  eV. A  $4 \times 4 \times 1$   $k$ -point mesh was used for electronic analysis.

The adsorption energy of the intermediate on the catalyst is defined as

$$E_{\text{ads}} = E_{\text{total}} - E_{\text{surface}} - E_{\text{int}} \quad (1)$$

where  $E_{\text{total}}$  is the energy of the Cu or Cu–IL catalyst adsorbing the intermediate,  $E_{\text{surface}}$  is the energy of the Cu surface or the Cu–IL catalyst, and  $E_{\text{int}}$  is the energy of the free intermediate. The formula for calculating the d-band center is as follows:<sup>46</sup>

$$\varepsilon_d = \frac{\int_{-\infty}^{\infty} \varepsilon \times \rho_d(\varepsilon) d\varepsilon}{\int_{-\infty}^{\infty} \rho_d(\varepsilon) d\varepsilon} \quad (2)$$

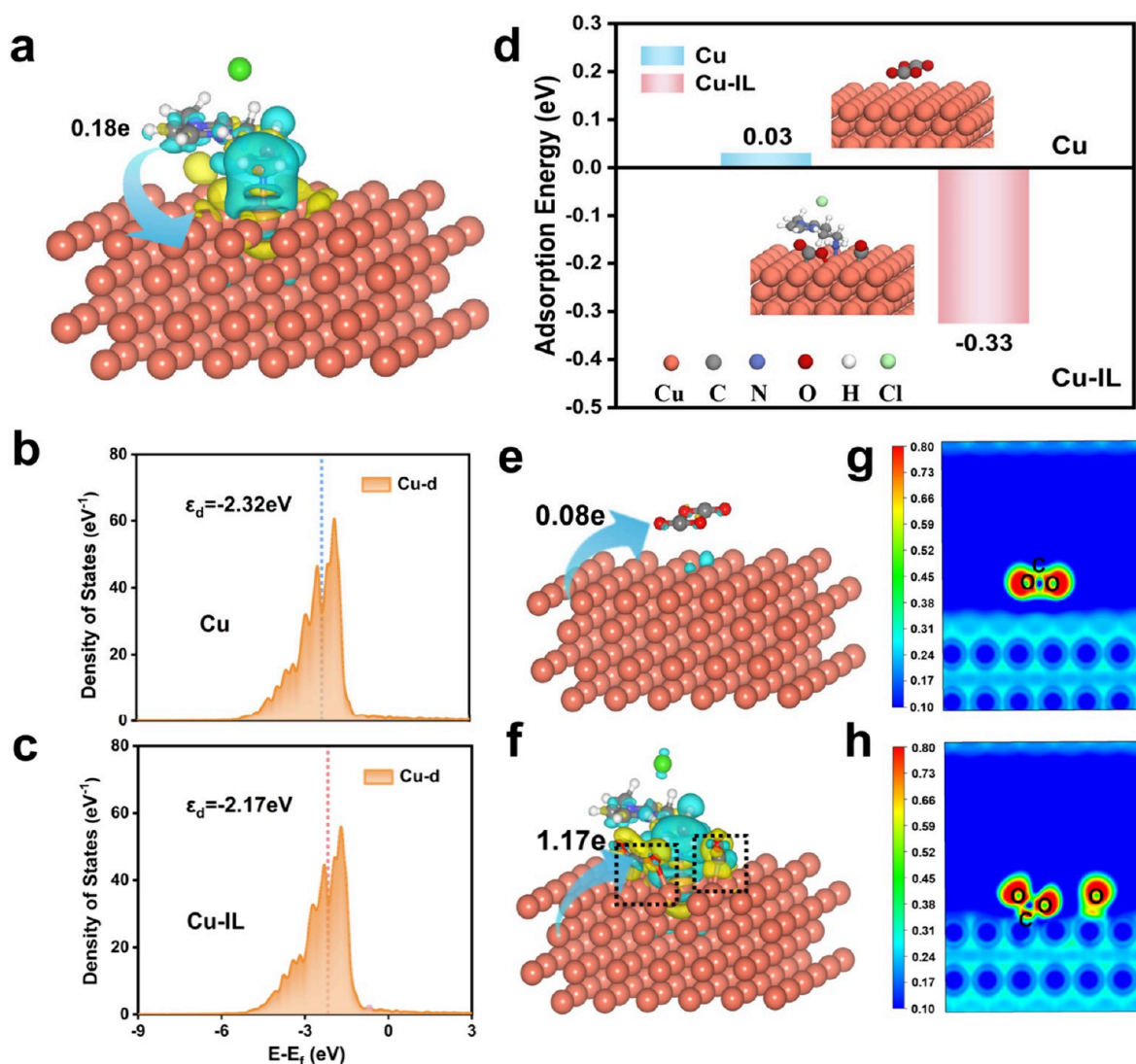
where  $\rho_d(\varepsilon)$  is the projected density of states of the d orbitals, representing the number of electronic states of the d orbitals at energy  $\varepsilon$ . The numerator is the integral of the product of the energy and the d state density over the entire energy range, and the denominator is the integral of the d state density over the entire energy range. The zero point of energy  $\varepsilon$  is usually chosen as the Fermi energy level. At this time,  $\varepsilon_d$  represents the position of the d-band center relative to the Fermi energy level. In this work, the VASPKIT package was employed to perform the extraction and analysis of the d-band center based on the projected density of states (PDOS) data.<sup>47</sup> The Gibbs free energy of formation ( $\Delta G$ ) at each step was calculated using the following formula:<sup>48</sup>

$$\Delta G = \Delta E + \Delta E_{\text{ZPE}} - T\Delta S \quad (3)$$

where  $\Delta E$  is the adsorption energy of the adsorbate,  $T$  is the temperature (set to 300 K), and  $\Delta E_{\text{ZPE}}$  and  $\Delta S$  are the differences in zero-point energy and zero-point entropy, respectively. For the transition state calculations, the nudged elastic band (NEB) method was employed.<sup>49,50</sup> The details of the electrocatalytic CO<sub>2</sub>RR experiments and related tests are provided in the [Supporting Information](#).

## RESULTS AND DISCUSSION

**Enhanced CO<sub>2</sub> Activation and Adsorption through the IL-Induced Confinement Effect.** Enhancing the efficiency of the CO<sub>2</sub> supply and improving its adsorption capability in the vicinity of catalysis are crucial for advancing the CO<sub>2</sub>RR process. [Figure 1](#) and [Figure S1](#) contrast the CO<sub>2</sub> adsorption conformations in the IL-modified Cu system versus the pristine Cu system. MD trajectories (2000 ps, 300 K) demonstrate that IL molecules undergo interfacial reconstruction, transitioning from a homogeneous aqueous dispersion to a stratified architecture with dual confinement effects on CO<sub>2</sub> ([Figure 1a–h](#)). A significant fraction of IL molecules chemisorb directly onto the Cu surface, forming a chemically anchored monolayer that enhances CO<sub>2</sub> chemisorption through a quantum confinement effect. The remaining IL constituents self-assemble into a secondary physisorption layer with thickness of  $2.4 \pm 0.1$  Å and stabilize  $2.6 \pm 0.2$  Å above the Cu surface. MSD analysis quantitatively demonstrates a remarkable suppression of CO<sub>2</sub> mobility in the Cu–IL system, with the diffusion coefficient reduced to  $0.409 \times 10^{-7}$  cm<sup>2</sup> s<sup>−1</sup>, nearly an order of magnitude lower than that of the IL-free counterpart ([Figure S2](#) and [Table S1](#)). This pronounced reduction in CO<sub>2</sub> diffusivity directly correlates with the IL-enhanced interfacial confinement effect. Density distribution profiles provide critical evidence for this confinement mechanism. Approximately 70% of CO<sub>2</sub> molecules accumulate within 1.74 Å above the Cu surface in the Cu–IL system, a characteristic distance range for strong chemisorption interactions between CO<sub>2</sub> and the Cu surface ([Figure S3](#)). The residue CO<sub>2</sub> molecules are effectively trapped within the secondary IL dynamic layer, as evidenced by a distinct secondary density peak at 2.74–3.29 Å, precisely corresponding to the spatial dimensions of the secondary IL physisorption layer. In contrast, a majority of the CO<sub>2</sub> molecules escape rapidly into the bulk solution for pristine Cu ([Figures S1–S3](#)). Consequently, the IL-induced confinement architecture



**Figure 2.** (a) Differential charge density of the Cu–IL surface. PDOS of several Cu atoms near  $\text{NH}_2$  for (b) pristine Cu and (c) Cu–IL. (d) Optimized adsorption energies of  $\text{CO}_2$  on the Cu and Cu–IL systems. Differential charge density of adsorption of  $\text{CO}_2$  on the (e) Cu and (f) Cu–IL systems. The cyan and yellow areas indicate charge depletion and accumulation, respectively. ELF maps of  $\text{CO}_2$  adsorption on the (g) Cu and (h) Cu–IL systems.

establishes a  $\text{CO}_2$ -enriched interfacial microenvironment through interfacial adsorption and spatiotemporal molecular trapping, which would facilitate subsequent activation and electron transfer processes to ultimately boost the  $\text{CO}_2\text{RR}$  and direct reaction pathways toward  $\text{C}_2$  products.

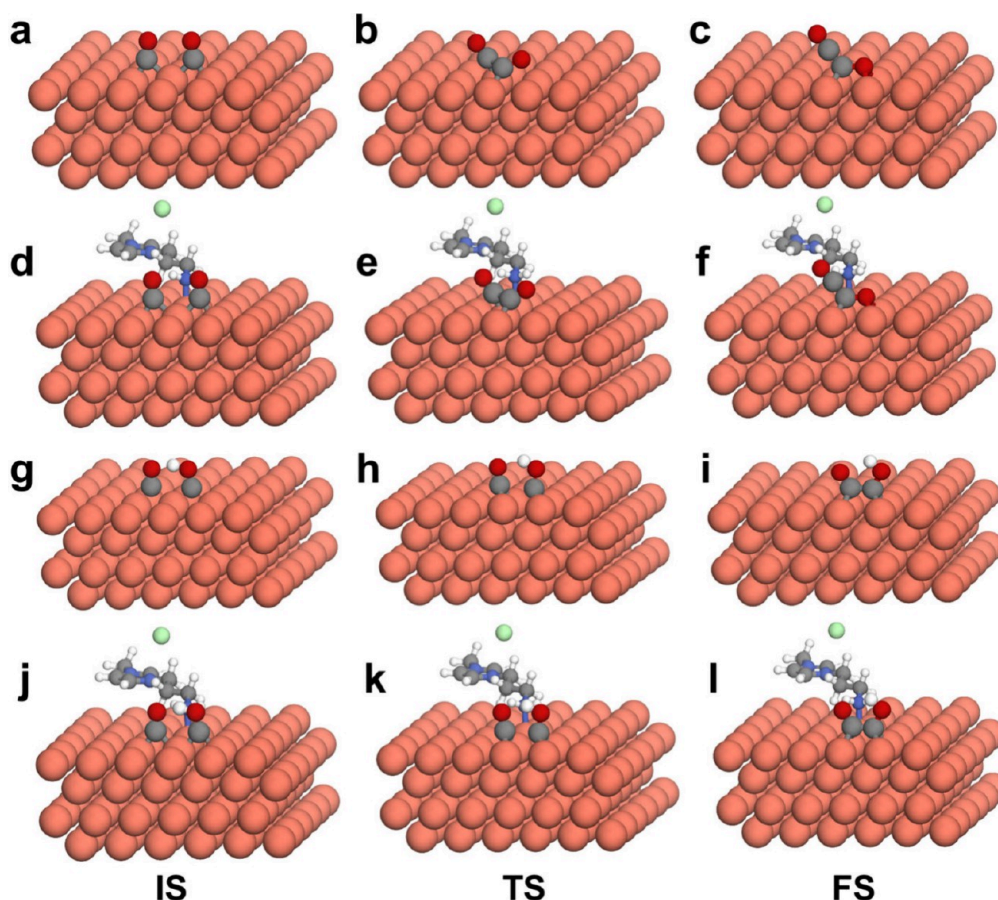
#### Influence of IL on the Electronic Properties for Cu.

Building on this interfacial confinement phenomenon, we further elucidate the electronic origins of enhanced  $\text{CO}_2$  activation through first-principles DFT calculations. Panels a and b of Figure S4 depict the configurations of pure Cu and IL-modified Cu substrates. Compared to bare Cu, a significant increase in Bader charge values is observed for copper atoms adjacent to the ILs, as shown in panels c and d of Figure S4. The differential charge density map (Figure 2a) reveals substantial electron transfer of 0.18 electron (e) from ILs to surface-proximal Cu atoms, inducing significant surface charge redistribution. This confirms the formation of electron-rich regions at Cu–IL interfaces, accompanied by an upward shift in the d-band center from  $-2.32$  eV for pristine Cu to  $-2.17$  eV for Cu–IL (Figure 2b,c). This upward shift of the d-band

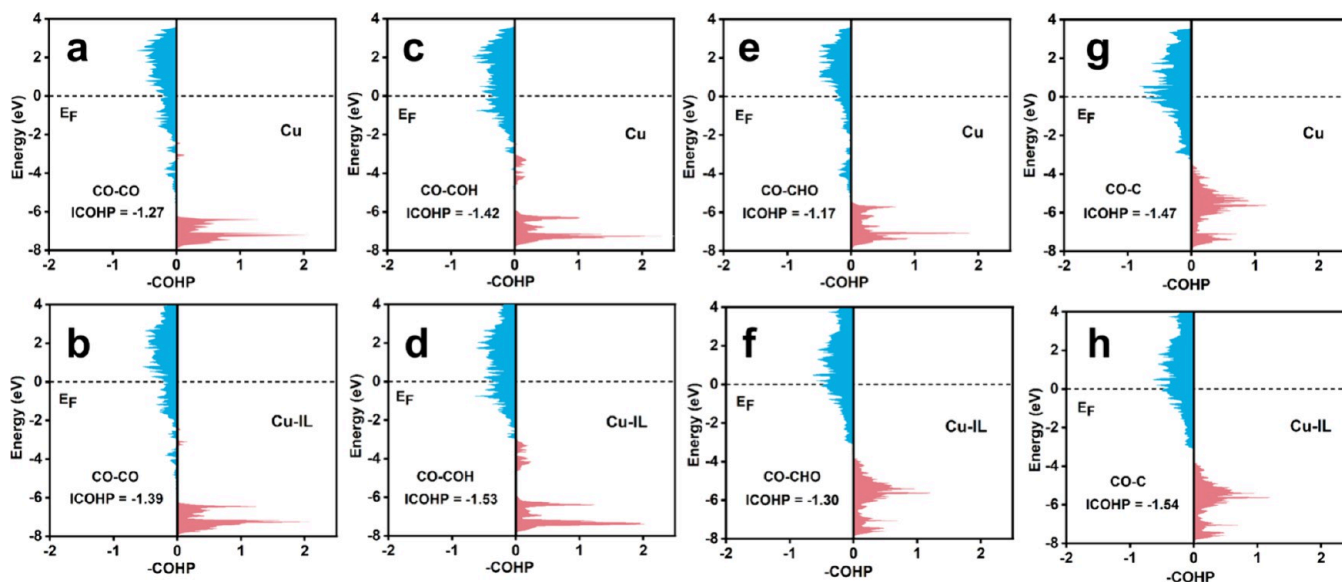
center reduces the population of antibonding states near the Fermi level, thereby enhancing the interaction between Cu and the adsorbates. Consequently, this substantial electron restructuring decreases the  $\text{CO}_2$  adsorption energy, as demonstrated by the transition from nearly neutral physisorption (0.03 eV) on pristine Cu to strong chemisorption ( $-0.33$  eV) on Cu–IL (Figure 2d and Figure S5). Furthermore, the differential charge density analysis quantifies a 14.6-fold enhancement in electron accumulation at adsorbed  $\text{CO}_2$ . The substantial enhancement from 0.08 e on pristine Cu to 1.17 e on the Cu–IL establishes a direct mechanistic correlation between IL-induced interfacial electronic restructuring and improved  $\text{CO}_2$  activation efficiency (Figure 2e,f). The ILs significantly modulate charge redistribution at the catalyst surface, thereby facilitating the critical initial step of  $\text{CO}_2$  reduction through enhanced electron donor–acceptor interactions.

The electron localization function (ELF) is particularly valuable in visualizing the electron density distribution and spatial localization. The red regions with higher ELF values





**Figure 3.** Optimized configurations of the IS, TS, and FS (side view): (a–c)  $^*\text{CO} + ^*\text{CO}$  on pristine Cu, (d–f)  $^*\text{CO} + ^*\text{CO}$  on Cu–IL, (g–i)  $^*\text{CO} + ^*\text{COH}$  on pristine Cu, and (j–l)  $^*\text{CO} + ^*\text{COH}$  on Cu–IL.



**Figure 4.** Projected COHP results of the overlap states between the  $^*\text{C}$  atom and the adsorbed Cu atoms in the initial state: (a and b)  $^*\text{CO} + ^*\text{CO}$  on pristine Cu and Cu–IL, (c and d)  $^*\text{CO} + ^*\text{COH}$  on pristine Cu and Cu–IL, (e and f)  $^*\text{CO} + ^*\text{CHO}$  on pristine Cu and Cu–IL, and (g and h)  $^*\text{CO} + ^*\text{C}$  on pristine Cu and Cu–IL, respectively.

present strong electron localization, typically associated with covalent bonding or lone pairs, while the blue regions signify weak electron localization and minimal bonding activity. As shown in panels g and h of Figure 2, the Cu–IL system exhibits pronounced electron localization at the  $\text{CO}_2$

adsorption sites, contrasting with delocalized electron distributions on pristine Cu. Such an IL-induced electronic distribution could facilitate directional orbital interactions between  $\text{CO}_2$  and Cu and promote both  $\text{CO}_2$  stabilization and subsequent activation for reduction pathways, particularly C–

C coupling steps that are critical for multicarbon product formation. These findings underscore a dual functionality of ILs during the initial steps of CO<sub>2</sub>RR catalysis: (i) creating a CO<sub>2</sub>-enriched microenvironment through an interfacial confinement effect and (ii) electronically activating both the catalyst and the reactant through charge transfer mechanisms, thereby promoting a more efficient catalytic process.

**Critical Role of the IL in Steering Reaction Mechanisms for CO<sub>2</sub> to C<sub>2</sub> Conversion.** *Effect of the IL on C–C Coupling.* The electrochemical reduction of the CO<sub>2</sub>RR to C<sub>2</sub> products is governed by two critical stages, the C–C coupling step and the post-C–C coupling step, which are regarded as the rate-determining step (RDS) and selectivity-determining step, respectively. It is essential for rational catalyst design to have a mechanistic understanding of how these steps are modulated by interfacial interactions. In this context, we systematically investigate the role of ILs in tuning the adsorption configurations and energetics of key intermediates during the CO<sub>2</sub>RR. Four plausible C–C coupling pathways have been identified and analyzed: \*CO + \*CO, \*CO + \*COH, \*CO + \*CHO, and \*CO + \*C. The optimized configurations of the initial state (IS), transition state (TS), and final state (FS) for these pathways are depicted in Figure 3 and Figures S6–S8. For the adsorption configurations in the \*CO + \*CO pathway, one adsorbate CO aligns parallel to the Cu interface and the other CO forms a C–C bond with its oxygen atom oriented toward the vacuum (Figure 3a–f and Figure S6a–f). In the \*CO + \*COH pathway, both intermediates bond to Cu via their carbon atoms, with the C–C bond parallel to the interface, and the oxygen atom of \*COH points toward the vacuum (Figure 3j–l and Figure S6j–l).

Similar analyses for the \*CO + \*CHO and \*CO + \*C pathways are provided in Figures S7 and S8, respectively. Although ILs induce no significant structural reconfiguration of adsorbed intermediates, distinct adsorption configurations are observed across different C–C coupling pathways. The adsorption site and the orientation of the adsorbed species are critical factors in determining the overall interaction strength between the intermediates and the catalyst surface, which ultimately determines the optimal energy pathways for the CO<sub>2</sub> reduction process.

The integrated crystal orbital Hamiltonian population (ICOHP) analyses<sup>51</sup> shown in Figure 4 demonstrate the electronic interactions between key intermediates and Cu surfaces during C–C coupling. On pristine Cu, the ICOHP values for \*CO + \*CO, \*CO + \*COH, \*CO + \*CHO, and \*CO + \*C are −1.27, −1.42, −1.17, and −1.47 eV, respectively, indicating moderate to strong Cu–C interactions. The incorporation of ILs strengthens these interactions, yielding more negative ICOHP values of −1.39, −1.53, −1.30, and −1.54 eV, respectively. This enhancement follows the sequence \*CO + \*CHO < \*CO + \*CO < \*CO + \*COH < \*CO + \*C, with \*CO + \*C exhibiting the most negative ICOHP value and strongest Cu–C interaction.

The integration of bond length analysis with the ICOHP results offers a comprehensive understanding of how ILs enhance C–C coupling by modulating both structural and electronic properties at the catalyst surface. Table 1 illustrates that the IL modifications consistently shorten the Cu–C bonds across all C–C coupling processes. For instance, in the \*CO + \*CO and \*CO + \*COH coupling reactions, Cu–C bond lengths at the IS decrease from 1.999 and 1.926 Å on the pure

**Table 1. Cu–C Bond Lengths (angstroms) and Distances from Cu to C**

| sample   | IS    |       | TS    |       | FS    |       |
|----------|-------|-------|-------|-------|-------|-------|
|          | Cu    | Cu–IL | Cu    | Cu–IL | Cu    | Cu–IL |
| *CO–*CO  | 1.997 | 1.986 | 1.974 | 1.971 | 2.025 | 2.024 |
| *CO–*COH | 1.926 | 1.919 | 2.000 | 1.970 | 2.043 | 2.041 |
| *CO–*CHO | 2.046 | 2.017 | 2.034 | 1.968 | 2.083 | 2.050 |
| *CO–*C   | 1.936 | 1.916 | 2.096 | 1.923 | 2.062 | 2.098 |

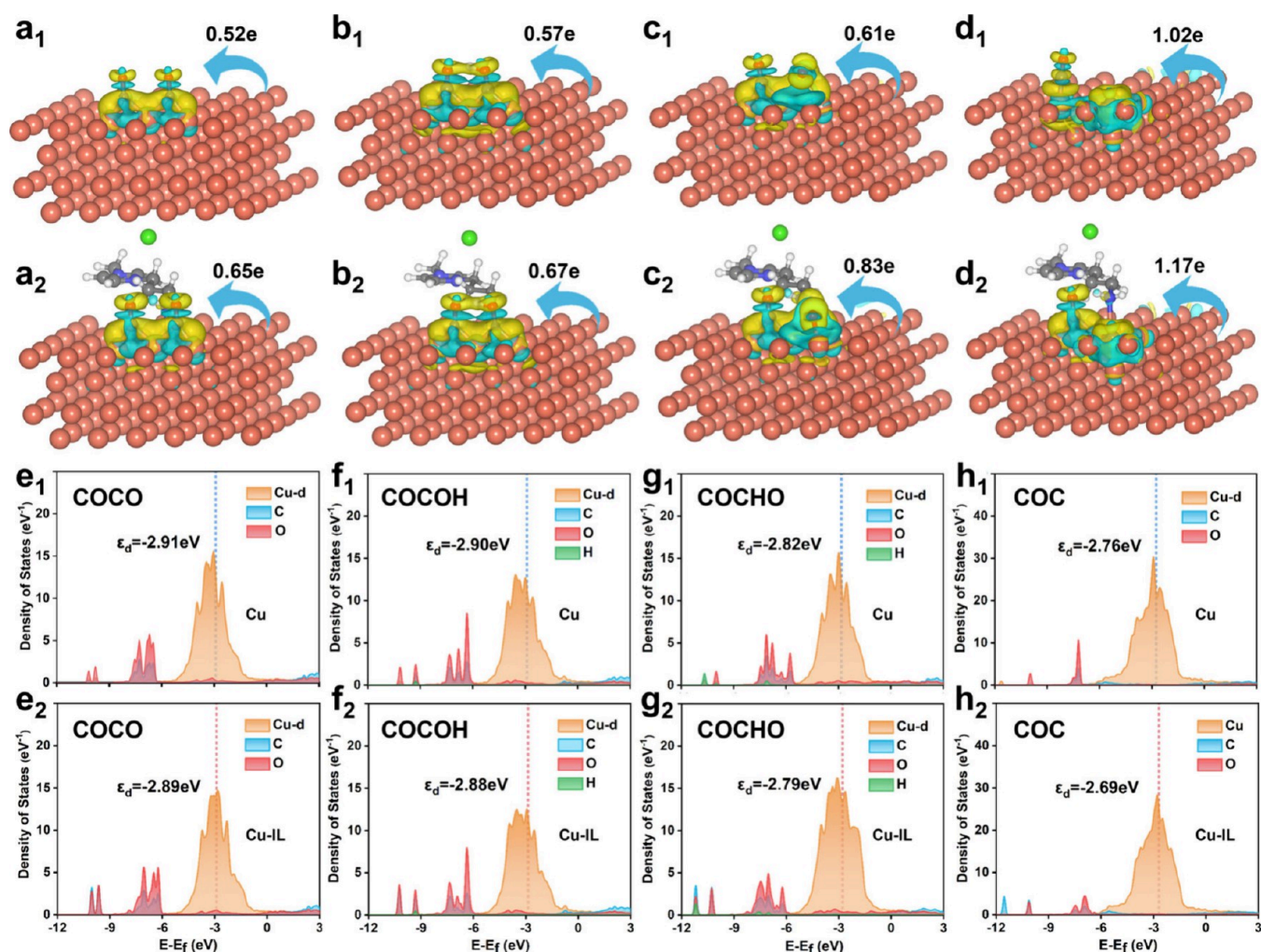
copper surface to 1.986 and 1.919 Å on Cu–IL, respectively. This structural contraction correlates with enhanced ICOHP values (for example, from −1.27 to −1.39 eV for \*CO + \*CO and from −1.42 to −1.53 eV for \*CO + \*COH upon incorporation of ILs). These trends indicate that ILs enhance the interaction between Cu and the intermediates by modifying the electron density at the Cu surface and optimizing the steric configurations. The shortened Cu–C bonds and strengthened electronic interactions synergistically stabilize intermediates and accelerate reaction kinetics. Specifically, the \*CO + \*C system exhibits the shortest Cu–C bond length (1.919 Å) and the most negative ICOHP value (−1.54 eV), indicating the strongest Cu–C interaction.

However, the doping of the C atom into the Cu substrate (Figures S7 and S8) would introduce steric and electronic complexities that may impede C–C coupling despite its favorable bonding strength. In contrast, the \*CO + \*COH pathway benefits from both strong Cu–C bonding (−1.53 eV) and favorable spatial alignment for C–C bond formation, rendering it a promising route for selective C<sub>2</sub> production. These findings highlight the necessity of balancing electronic and structural factors when designing IL-modified catalysts.

To further elucidate the intrinsic mechanisms of IL-mediated C–C coupling enhancement during the CO<sub>2</sub>RR, we systematically investigated the electronic structure evolution of key intermediates on pristine and IL-modified Cu surfaces. Differential charge density analyses demonstrate that IL functionalization triggers pronounced interfacial charge transfer and redistribution between the intermediates and the Cu surface (Figure 5a<sub>1</sub>–d<sub>2</sub>). For the \*CO + \*COH coupling mode, the Cu–IL surface exhibits a charge depletion of approximately 0.67 e (compared to 0.57 e on pristine Cu), with transferred electrons predominantly accumulating at the \*COCO intermediate.

This IL-induced increase in charge accumulation correlates with an improved stabilization of intermediates, as confirmed by characteristic shifts in the d-band center toward the Fermi level (Figure 5e<sub>1</sub>–h<sub>2</sub>). For instance, the d-band center for \*CO + \*COH shifts from −2.90 eV (pristine Cu) to −2.88 eV (Cu–IL), indicating stronger orbital hybridization between adsorbates and the modified surface in the Cu–IL system. Notably, the observed enhanced overlap between Cu d orbitals and C/O p orbitals directly facilitates C–C bond formation by reducing activation barriers through an improved orbital interaction. This stabilization effect extends universally across all investigated coupling pathways, demonstrating the general efficacy of IL modification. The d-band centers shift from −2.91 to −2.89 eV for \*CO + \*CO, from −2.82 to −2.79 eV for \*CO + \*CHO, and from −2.76 to −2.69 eV for \*CO + \*C. The systematic upshift of d-band centers toward the Fermi level consistently follows the d-band center theory, where higher-level d states strengthen adsorbate–surface interactions through enhanced electron back-donation. Our findings





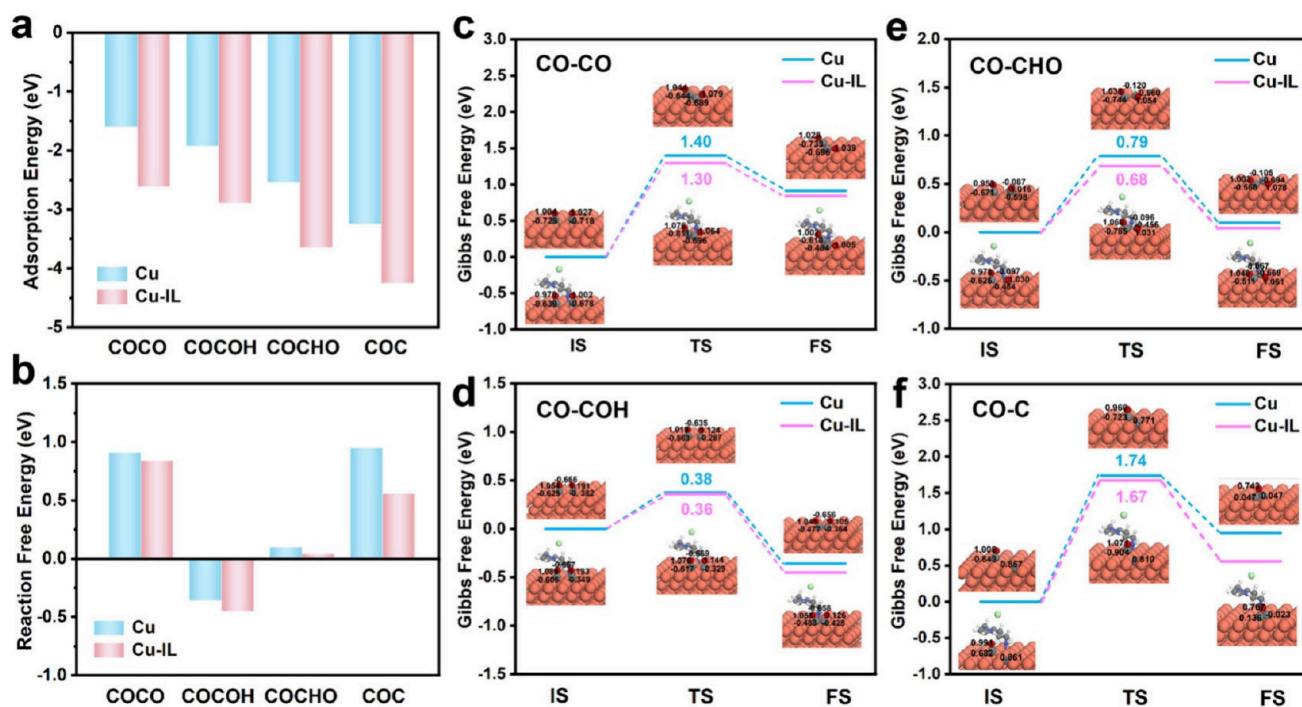
**Figure 5.** (a–d) Calculated differential charge densities of various proposed intermediates on the Cu and Cu–IL systems, with cyan and yellow areas representing charge depletion and charge accumulation, respectively. (e–h) DOS plots for  $\ast\text{CO} + \ast\text{CO}$ ,  $\ast\text{CO} + \ast\text{COH}$ ,  $\ast\text{CO} + \ast\text{CHO}$ , and  $\ast\text{CO} + \ast\text{C}$ , respectively, adsorbed on pristine Cu and Cu–IL systems.

establish IL modification as an effective strategy for intermediate stabilization through three atomic-level mechanisms: (i) enhanced interfacial charge transfer, (ii) optimized d-band positioning near the Fermi level, and (iii) improved orbital overlap for bonding interactions.

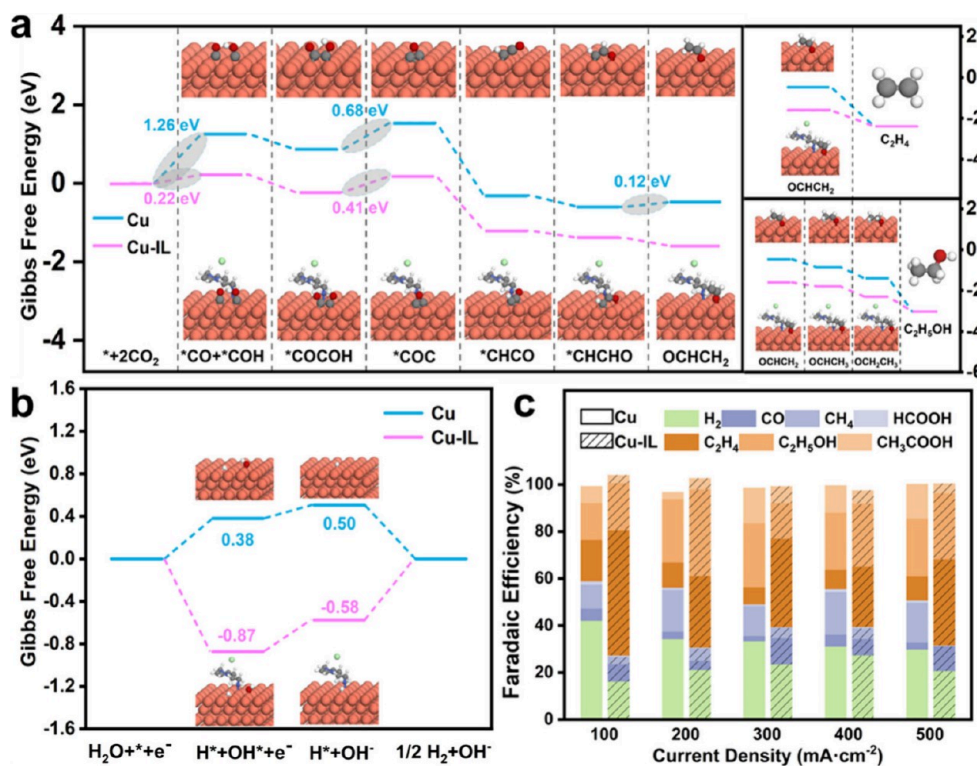
**IL-Engineered Cu Catalysts for Electrocatalytic  $\text{CO}_2$  to  $\text{C}_2$  Conversion: Thermodynamic–Kinetic Synergy and Mechanistic Insights.** Comprehensive DFT calculations from an energy perspective were performed to systematically investigate the C–C coupling mechanisms on the IL-modified Cu surface. The adsorption energies (Figure 6a) of the dimer configurations of key intermediates decrease in the following order:  $\ast\text{CO} + \ast\text{CO}$  ( $-2.61\text{ eV}$ ) >  $\ast\text{CO} + \ast\text{COH}$  ( $-2.89\text{ eV}$ ) >  $\ast\text{CO} + \ast\text{CHO}$  ( $-3.65\text{ eV}$ ) >  $\ast\text{CO} + \ast\text{C}$  ( $-4.25\text{ eV}$ ). This demonstrates 25–40% stronger adsorption compared to pristine Cu. This systematic enhancement directly aligns with upward shifts in the d-band center positions toward the Fermi level. Generally, according to Sabatier-optimal adsorption energies, it should critically balance two competing requirements: sufficient stabilization of intermediates to enable C–C bond formation while avoiding excessive binding that would impede product desorption. This balance manifests most prominently in the  $\ast\text{CO} + \ast\text{COH}$  configuration ( $-2.89\text{ eV}$ ), which combines a moderate adsorption strength with favorable

dimerization thermodynamics. Complementary reaction energy analysis (Figure 6b) reveals that the  $\ast\text{CO} + \ast\text{COH} \rightarrow \ast\text{CO}-\ast\text{COH}$  pathway possesses the most thermodynamically favorable profile with the lowest reaction energies of  $-0.36\text{ eV}$  (pristine Cu) and  $-0.45\text{ eV}$  (Cu–IL), making it exothermic and thermodynamically spontaneous.

This thermodynamic advantage in IL-modified systems was further amplified by kinetic studies through NEB calculations. Figure 6c–f depicts the corresponding free energy profiles and minimum energy paths (MEP), where IL modification systematically lowers activation barriers across all pathways: from 1.40 to 1.30 eV for  $\ast\text{CO} + \ast\text{CO}$ , from 0.38 to 0.36 eV for  $\ast\text{CO} + \ast\text{COH}$ , from 0.79 to 0.68 eV for  $\ast\text{CO} + \ast\text{CHO}$ , and from 1.74 to 1.67 eV for  $\ast\text{C} + \ast\text{CO}$ . The  $\ast\text{CO} + \ast\text{COH}$  pathway emerges as the kinetically dominant one, combining the lowest activation barrier (0.36 eV) with optimal thermodynamic stability. For the reaction pathway, the C–C bond length and the interatomic distances between C atoms from the initial state through the transition state to the final state are summarized in Table S2. Tables S3–S10 show the frequency vibrations and imaginary frequencies information for each transition state. The shortened C–C bond lengths and weakened interatomic distances also confirm that the IL promotes C–C coupling. A dual advantage arises from IL-



**Figure 6.** (a) Adsorption energies and (b) reaction free energies of \*COCO, \*COCO, \*COCHO, and \*COC intermediates on the Cu and Cu-IL systems. (c–f) Gibbs free energies of the \*COCO, \*COCO, \*COCHO, and \*COC intermediates, respectively, on the Cu and Cu-IL systems.



**Figure 7.** (a) Free energy diagram of the CO<sub>2</sub>RR on the pristine Cu and Cu-IL systems. The insets are the corresponding configurations of key intermediates. (b) Free energy diagram of the HER for the pristine systems. (c) Faradaic efficiency of different products produced on Cu and Cu-IL catalysts at different currents.

induced charge redistribution: simultaneous stabilization of transition states and reduction of the reaction energy. This synergistic effect underscores the ILs' role in modifying both the thermodynamic profile and the kinetic trajectory of the

RDS. Crucially, the IL microenvironment induces charge redistribution as evidenced by Bader analysis (insets of Figure 6c–f), where enhanced electron transfer from Cu to \*C/O-containing intermediates strengthens metal–adsorbate inter-



actions. The modified electronic structure facilitates stronger orbital hybridization between Cu surface states and intermediate frontier orbitals, thereby enhancing intermediate stabilization, which is a critical prerequisite for efficient C–C coupling.

Subsequently, critical thermodynamic and kinetic advantages of IL-modified Cu catalysts for the electrocatalytic CO<sub>2</sub>RR to yield C<sub>2</sub> products (C<sub>2</sub>H<sub>4</sub> and C<sub>2</sub>H<sub>5</sub>OH) were confirmed through combined DFT calculations, in situ spectroscopic analysis, and electrochemical validation. The Gibbs free energy diagram in Figure 7a reveals a profound restructuring of reaction thermodynamics upon IL functionalization, with the initial CO<sub>2</sub> activation step exhibiting a decrease in the change in free energy ( $\Delta G$ ) from 1.26 eV (pristine Cu) to 0.22 eV (Cu–IL). This thermodynamic advantage extends coherently across the subsequent reaction steps, as ILs stabilize key intermediates and decrease the thermodynamic energy during the CO<sub>2</sub>RR, thereby significantly enhancing the overall reaction efficiency. In particular, the free energy difference for the \*COCO<sub>2</sub>H intermediates is a decisive factor in steering the reaction pathway toward C<sub>2</sub> product formation, with a significant decrease (from 0.68 eV on pure Cu to 0.41 eV on the Cu–IL surface) noted upon the incorporation of ILs. Moreover, the persistent negative free energy profile along the CO<sub>2</sub>RR pathway after this RDS confirms the thermodynamic favorability for C<sub>2</sub> product formation in the Cu–IL system.

Direct experimental validation was achieved through in situ attenuated total reflection–Fourier transform infrared (ATR–FTIR) spectroscopy (Figure S9) conducted during the CO<sub>2</sub>RR.

The Cu–IL catalyst exhibits characteristic stretching vibrational modes of adsorbed \*COOH at 1390 cm<sup>−1</sup> and asymmetric stretching vibrations of CO<sub>2</sub> at 2360 and 2331 cm<sup>−1</sup> and CHO at 1030 cm<sup>−1</sup>, aligning precisely with DFT-predicted intermediates. The emergence of the COCO<sub>2</sub>H dimer peak at 1260 cm<sup>−1</sup> confirms the IL-mediated stabilization of this C–C coupling intermediate.<sup>52,53</sup> The spectroscopic identification of key intermediates provides direct experimental validation of the theoretically predicted stabilization effect on the CO<sub>2</sub>RR, establishing a molecular-level bridge between the thermodynamic framework derived from free energy diagrams and real-time catalytic behavior.

For the CO<sub>2</sub>RR, the HER as a detrimental competing process can diminish the efficiency of C<sub>2</sub> production. The IL-induced microenvironment exerts dual functionality in simultaneously promoting C<sub>2</sub> pathways and suppressing competing hydrogen evolution. The Gibbs free energy stepwise diagram for the HER (Figure 7b) reveals that incorporation of ILs promotes the dissociation of water into adsorbed hydrogen (\*H) and hydroxide (\*OH) species on the Cu surface, with a  $\Delta G$  of −0.87 eV for \*H/\*OH generation in comparison with a value of 0.38 eV on bare Cu. However, it simultaneously imposes a kinetic bottleneck for \*H desorption, as evidenced by an increased energy barrier of 0.58 eV for H removal (compared to −0.50 eV on pristine Cu). The action thermodynamically favoring \*H/\*OH generation while hindering \*H release arises from IL-induced electric modulation and proton confinement. This dual mechanism is quantitatively reflected in the dramatic Faradaic efficiency improvements at 100 mA cm<sup>−2</sup>. C<sub>2</sub>H<sub>4</sub> production surges from 17.59% to 53.27%, while C<sub>2</sub>H<sub>5</sub>OH production increases from 15.73% to 20.2%, accompanied by HER suppression from 41.96% to 16.11% (Figure 7c and Figure S10). Stability tests

demonstrated that the electrode potential and the FE of C<sub>2</sub> and C<sub>2</sub>H<sub>4</sub> remained stable over 12 h, indicating excellent electrochemical durability. Additionally, variations in cation type, ionic liquid concentration, and reaction temperature were systematically evaluated to confirm the stability and adaptability of the ILs in optimizing the CO<sub>2</sub>RR pathway (Figures S11–S14).

The divergent enhancement between C<sub>2</sub>H<sub>4</sub> and C<sub>2</sub>H<sub>5</sub>OH is rationalized through bond evolution analysis (Table S11). IL-induced shortening of Cu–O bonds enhances oxygenate retention, while elongated C–O bonds favor C–O cleavage, directing \*CHCH<sub>2</sub> intermediates toward ethylene via hydrogenation rather than ethanol via oxygen retention. This mechanistic nuance underscores the precision of IL-mediated microenvironment engineering in controlling the reaction trajectories. The integration of computational and experimental results provides critical insights into designing advanced catalytic interfaces by leveraging electrolyte-mediated electronic effects to control reaction pathways at the quantum level.

## CONCLUSIONS

This study unravels the molecular-level mechanism of the catalytic superiority of 1-(3-aminopropyl)-3-methylimidazole chloride (IL)-modified Cu electrodes in steering CO<sub>2</sub> electroreduction toward C<sub>2</sub> products. This mechanistic investigation holds significant implications for practical CO<sub>2</sub>RR systems. By reducing the RDS energy barrier through simultaneous thermodynamic stabilization and kinetic facilitation, IL modification enables operation at lower overpotentials while maintaining a high C<sub>2</sub> production efficiency. The convergence of adsorption energy trends, d-band center shifts, and activation barrier reductions provides compelling evidence that IL modification operates through electronic structure modulation rather than mere physical confinement effects. The strengthened Cu–\*C/O interactions directly correlate with enhanced C–C coupling efficiency. Through integrated DFT calculations, in situ ATR–FTIR spectroscopy, and electrochemical analysis, we demonstrate that IL functionalization induces a dual thermodynamic–kinetic synergy: (i) a drastic reduction of Gibbs free energy barriers for CO<sub>2</sub> activation and C–C coupling and (ii) profound suppression of the HER via proton confinement and electric modulation effects. The study of IL-mediated optimization of electrocatalytic interfaces proposes a universal and promising concept for designing high-selectivity, energy-efficient systems for sustainable carbon deployment.

## ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.5c02334>.

Electrode preparation method; experimental details; qualitative and quantitative analysis of reaction products by gas chromatography (GC) and nuclear magnetic resonance (NMR); analysis of the intermediate of the reduction reaction by Fourier transform infrared spectrometry; kinetic snapshots of the evolution of CO<sub>2</sub> adsorption and diffusion with time in the Cu system; MSD and diffusion coefficient of CO<sub>2</sub> diffusion on the surface of the original Cu and IL-modified Cu systems; distribution of CO<sub>2</sub> density in the vertical

direction at the interface between the original Cu and Cu–IL; configurations of the original Cu and Cu–IL systems before and after CO<sub>2</sub> adsorption, as well as the optimized configurations of the IS, TS, and FS under different reaction conditions (\*CO + \*CO, \*CO + \*COH, \*CO + \*CHO, and \*CO + \*C); C–C bond lengths or distances between C atoms in the original Cu and Cu–IL systems under different reaction conditions (COCO, COCOH, COCHO, and COC); frequency vibrations and imaginary frequency information about each transition state on the Cu and Cu–IL catalysts; results from in situ attenuated total reflection–Fourier transform infrared spectroscopy of the Cu–IL system; nuclear magnetic resonance spectrum of the Cu–IL system; Cu–O and C–O bond lengths in the \*OCHCH<sub>2</sub> state; electrocatalytic stability test with a long-term cycle of 12 h at a current density of 100 mA cm<sup>−2</sup>; Faradaic efficiency of different products produced on the Cu–IL (COOH) catalyst at different currents; Faradaic efficiencies of different products produced by the Cu–IL catalyst at IL concentrations of 0.001 and 0.01 M; and Faradaic efficiencies of different products produced by the Cu–IL catalyst at 50 and 75 °C (PDF)

## AUTHOR INFORMATION

### Corresponding Authors

**Congyun Zhang** – School of Environment and Geography, Qingdao University, Qingdao 266071, China; [orcid.org/0000-0001-7150-5602](https://orcid.org/0000-0001-7150-5602); Email: [zhangcy@qdu.edu.cn](mailto:zhangcy@qdu.edu.cn)

**Lijuan Shi** – School of Chemical Engineering and Pharmacy, Wuhan Institute of Technology, Wuhan 430205, China; [orcid.org/0000-0001-5829-0766](https://orcid.org/0000-0001-5829-0766); Email: [shilijuanwit@sina.com](mailto:shilijuanwit@sina.com)

### Authors

**Chuanhui Wang** – School of Environment and Geography, Qingdao University, Qingdao 266071, China

**Mingyue Qiu** – School of Chemical Engineering and Pharmacy, Wuhan Institute of Technology, Wuhan 430205, China

**Guizeng Liang** – School of Environment and Geography, Qingdao University, Qingdao 266071, China

**Hui Yan** – School of Pharmaceutical Sciences, Liaocheng University, Liaocheng, Shandong 252059, China; [orcid.org/0000-0002-9843-0601](https://orcid.org/0000-0002-9843-0601)

**Jiamin Ma** – School of Environment and Geography, Qingdao University, Qingdao 266071, China

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.langmuir.5c02334>

### Notes

The authors declare no competing financial interest.

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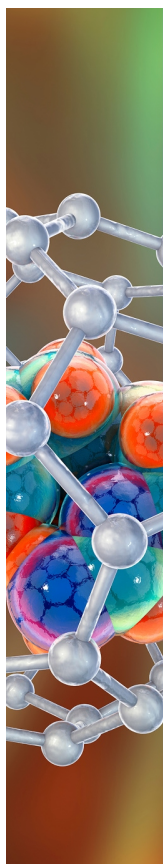
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A vertical graphic on the left side of the advertisement showing a complex molecular structure with various colored spheres (red, blue, green, orange) representing atoms, connected by grey rods representing bonds. The background of the entire advertisement is dark blue.

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